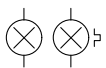
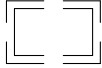
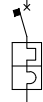
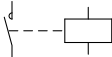
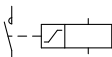
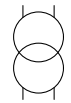
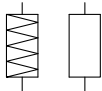


1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS																				
SAFETY CRITICAL ELEMENT LIST																				
IFI	2	31/01/2025	AS-BUILT				ADF	GDP	IC											
IFI	1	06/12/2024	GENEERAL REVIEW				ADF	GDP	IC											
IFI	0	26/03/2024	ISSUED FOR COMMENTs				ADF	GDP	IC											
Status	Rev. n.	Rev. n.	Rev. n.				Rev. n.	Rev. n.	Rev. n.											
	Rev. Index	Date	Description				Prepared	Checked	Approved											
Client							Project Title													
ManaEnergy Managem Group MANAENERGY Mahaj Riad Center, Batiment 6, 3eme etage Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco							TENDRARA PROJECT													
Contractor							Provision of a m-LNG plant for production for ManaEnergy													
ITALFLUID ITALFLUID GEOENERGY Marocco Branch Boulevard de Bourgogne Rue Jaafar Ibn Habib Residence Al Mashrik II 1ere Etage N3 Casablanca																				
Engineering																				
ITALFLUID ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) Tel:085-4224374 / fax: 085-4220742																				
Facility Name			Location				Scale		Sheet of Sheets											
TENDRARA			MOROCCO				A4		1 of 18											
Document Title					Doc. Code															
P-PK-PK-341-3					ELD01-ING-FR01															
AMMONIA COMPRESSOR PACKAGE 3					Vendor Doc. code															
					IE24090_SE00_PCS															
			Project :			Customer :			Object :											
			P-PK-PK-341-3			ManaEnergy Managem Group														
General Contractor :		Manufacture :		AMMONIA COMPRESSOR PACKAGE 3			Contract :			Rev. 2			Data : 31/01/2025			Date : 26/03/2024				
		esa esaelettromeccanica.com					S/N :			Rev. 1			Data : 06/12/2024			Sheet : 1		of : 18		

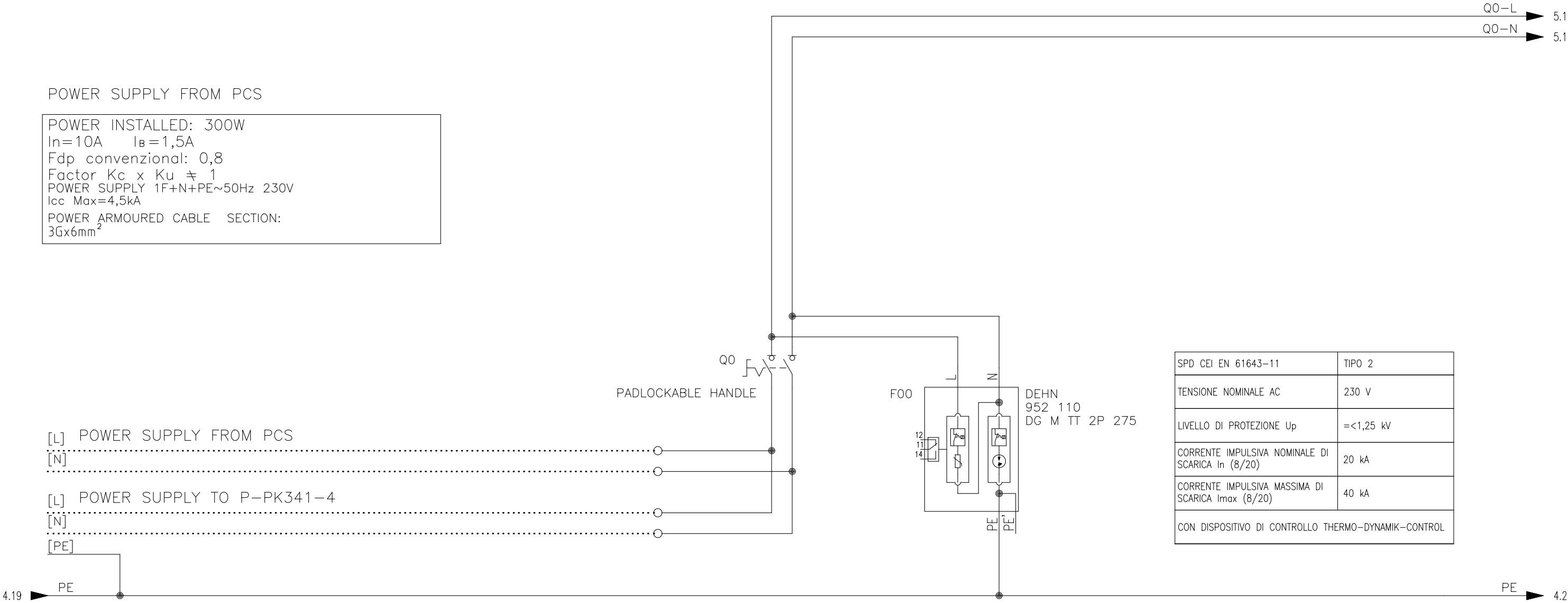
CIRCUIT SYMBOLS ACCORDING TO CEI STANDARD

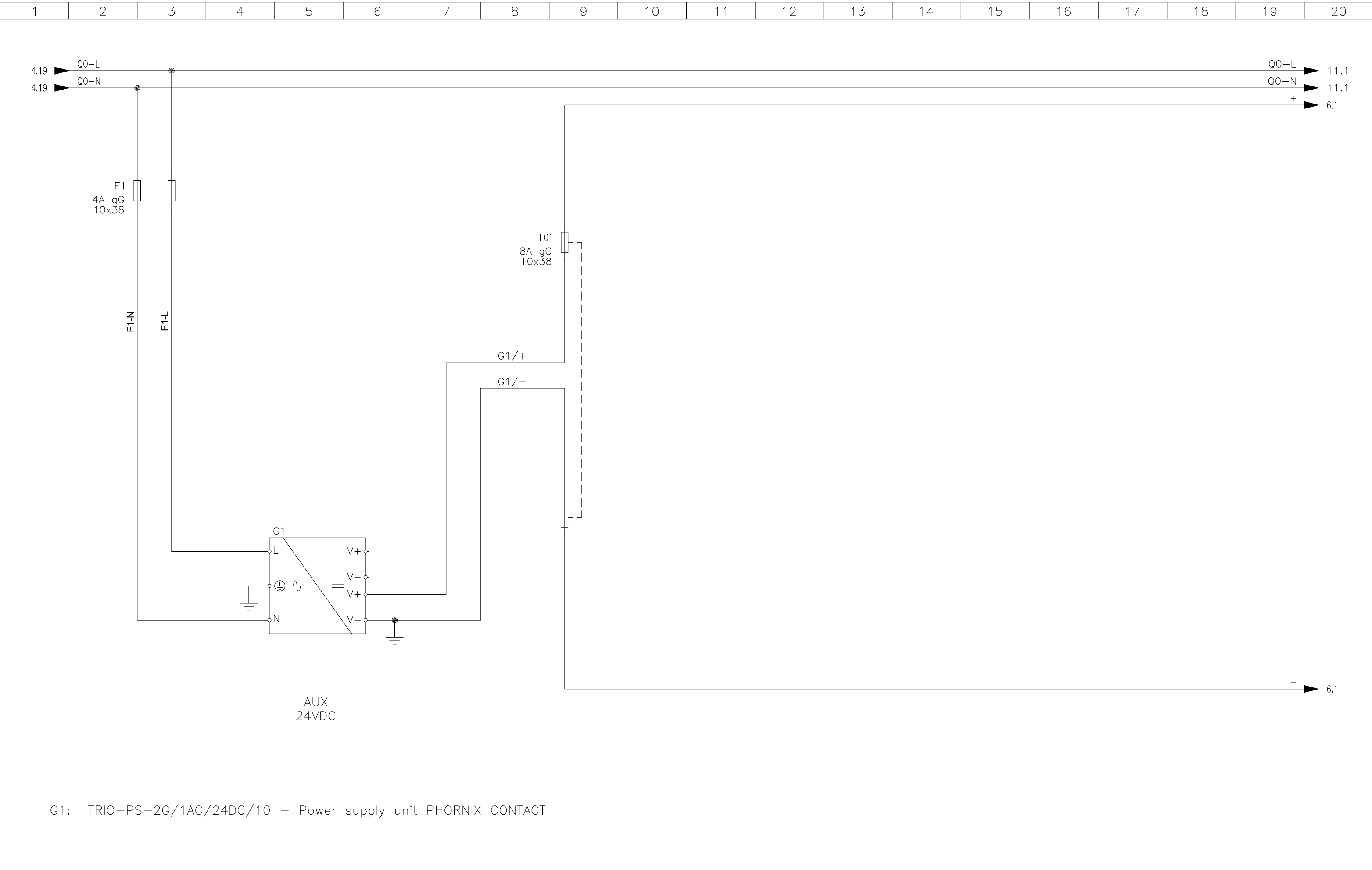
	ISOLATOR		ELECTRONIC TIMER WITH PICK-UP DELAY		MUSHROOM PUSH-BUTTON PULSE CONTACT WITH STAY-PUT MAKE CONTACT OPERATED BY STRIKING		STEDY OR FLASHING LIGHT
	FUSED ISOLATOR		ELECTRIC PULSE METER		KEY OPERATING DEVICE FOR SELECTORS, PUSH-BUTTON, ETC.		PLC UNIT
	CIRCUIT BREAKER WITH THERMAL AED MAGNETIC OVERCURRENT RELEASES		FUSE CARRIER		PRESSURE SWITCH		CONTROL INSTRUMENT
	CIRCUIT BREAKER WITH MAGNETIC RELEASE		MAKE CONTACT OF CONTACTOR		TEMPERATURE SWITCH		ENERGY METER
	CIRCUIT BREAKER WITH THERMAL, MAGNETIC AND DIFFERENTIAL RELEASES		MAKE CONTACT OF CONTACTOR		FLOW SWITCH		MEASURING INSTRUMENT
	RESIDUAL CURRENT CIRCUIT BREAKER "AC" TYPE FOR ALTERNATING CURRENT "A" TYPE FOR ALTERNATING OR PULSATING CURRENTS		MAKE AND BREAK CONTACT		FLOAT SWITCH		VOLTAGE TRANSFORMER
	THERMAL RELAY (OVERLOAD RELAY)		MAKE AND BREAK CONTACT WITH DROP-OUT DELAY		BUCHHOLZ RELAY		CURRENT TRANSFORMER
	COIL OF CONTACTOR RELAY		MAKE AND BREAK CONTACT WITH PICK-UP DELAY		CAM OPERATED DEVICE		ACOUSTIC SIGNALLING DEVICE
	COIL OF SHUT-OFF ALARM MAKE CONTACT ON ENERGIZATION		MAKE AND BREAK CONTACT WITH DROP-OUT AND PICK-UP DELAY (FLIP-FLOP)		PROXIMITY SWITCH		RESISTANCE
	COIL OF SHUT-OFF ALARM DENERGIZES IN CASE OF UNDERVOLTAGE		MAKE AND BREAK CONTACT OF THERMAL RELAY		TEMPERATURE SENSOR		CAPACITOR
	COIL OF OPERATING DEVICE WITH DROP-OUT DELAY		MAKE AND BREAK CONTACT OF LIMIT SWITCH		PHOTOELECTRIC SENSOR		SINGLE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH PICK-UP DELAY		MULTI POSITION SWITCH (GENERAL SYMBOL)		SOLENOID VALVE		THREE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH DROP-OUT AND PICK-UP DELAYS		MAKE AND BREAK CONTACT OPERATED BY TURNING WITH SPRING RETURN		PHOTOCELL		
	COIL OF LATCHING RELAY (STEP-BY-STEP FUNCTION)		MAKE AND BREAK CONTACT OPERATED BY PUSHING WITH SPRING RETURN				
	ELECTRONIC OR ELECTRO-MECHANICAL TIME SWITCH		MAKE AND BREAK CONTACT OPERATED BY PULLING WITH OR WITHUOT SPRING RETURN				

CONDUCTOR COLOR CODING INSIDE CONTROL PANEL	
AC AND DC POWER CIRCUIT	BLACK
NEUTRAL CONDUCTOR OF POWER CIRCUIT	LIGHT–BLUE
PROTECTIVE CONDUCTOR (EARTH CONDUCTOR)	YELLOW–GREEN
110/220V AC AUXILIARY CIRCUIT (*)	RED
24V AC AUXILIARY CIRCUIT	WHITE
24V AC AND DC PLC INPUT	GRAY
24V AC PLC OUTPUT	WHITE
24V AC AND DC INTERLOCKING CIRCUIT	ORANGE
24V DC AUXILIARY CIRCUIT AND PLC OUTPUT	DARK–BLUE
INTRINSIC SAFE SIGNAL	BLUE
(*) : RED FOR 110V AND BROWN FOR 220V, IN CASE BOTH VOLTAGE ARE USED	

CODE COLOUR	
COLOUR	CODE LETTERS
BLACK	BK
BROWN	BN
RED	RD
ORANGE	OG
YELLOW	YE
GREEN	GN
BLUE	BU
GREY	GY
WHITE	WH
SILVER	SR
CLEAR	CL
GREEN AND YELLOW	GNYE

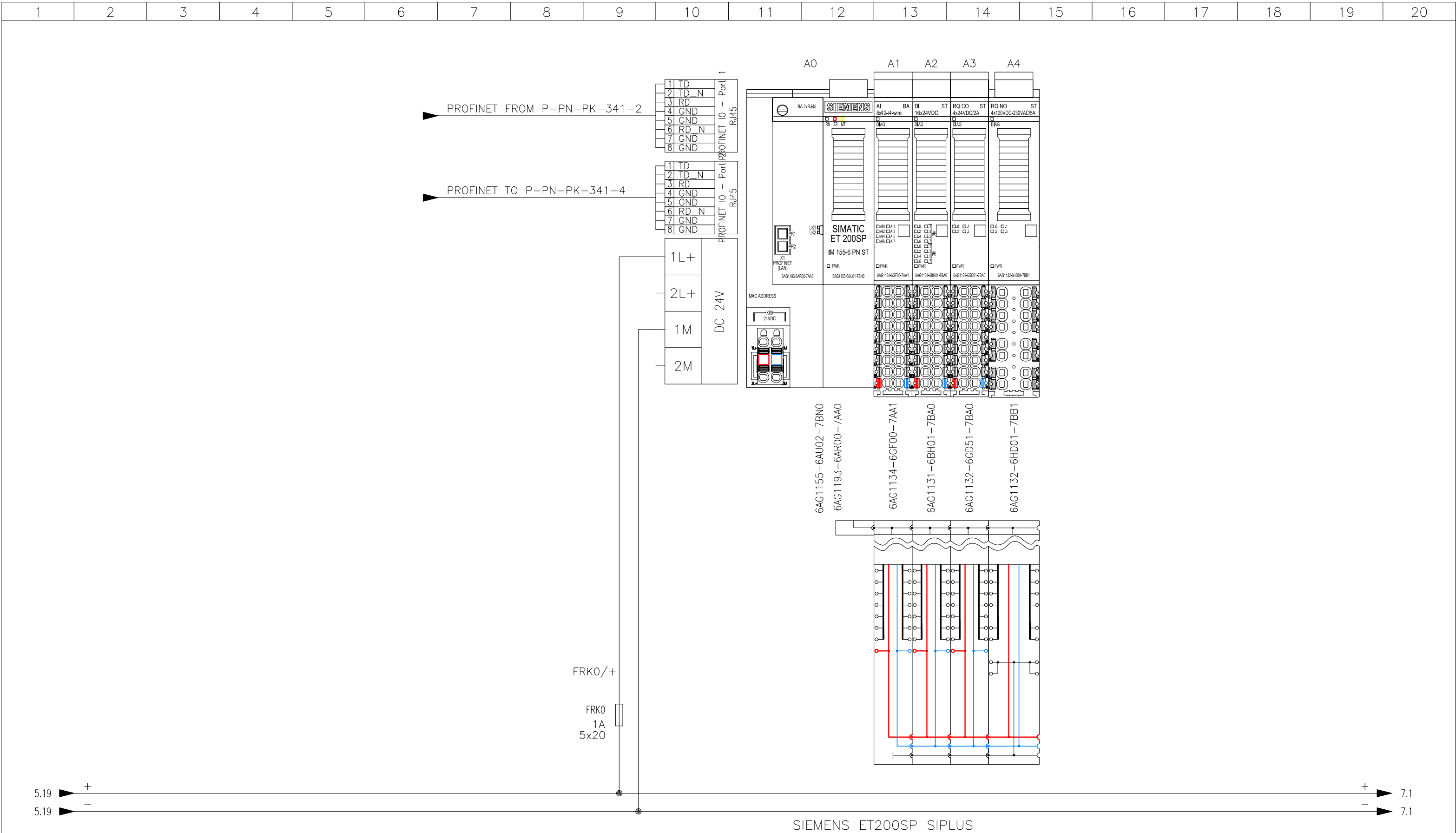
	POWER MAIN DISCONNECT SWITCH																	FEEDER ID	
	1F+N+PE~50Hz 230V																	RATED POWER Pn-RATED VOLTAGE Vn	
																		RATED CURRENT	

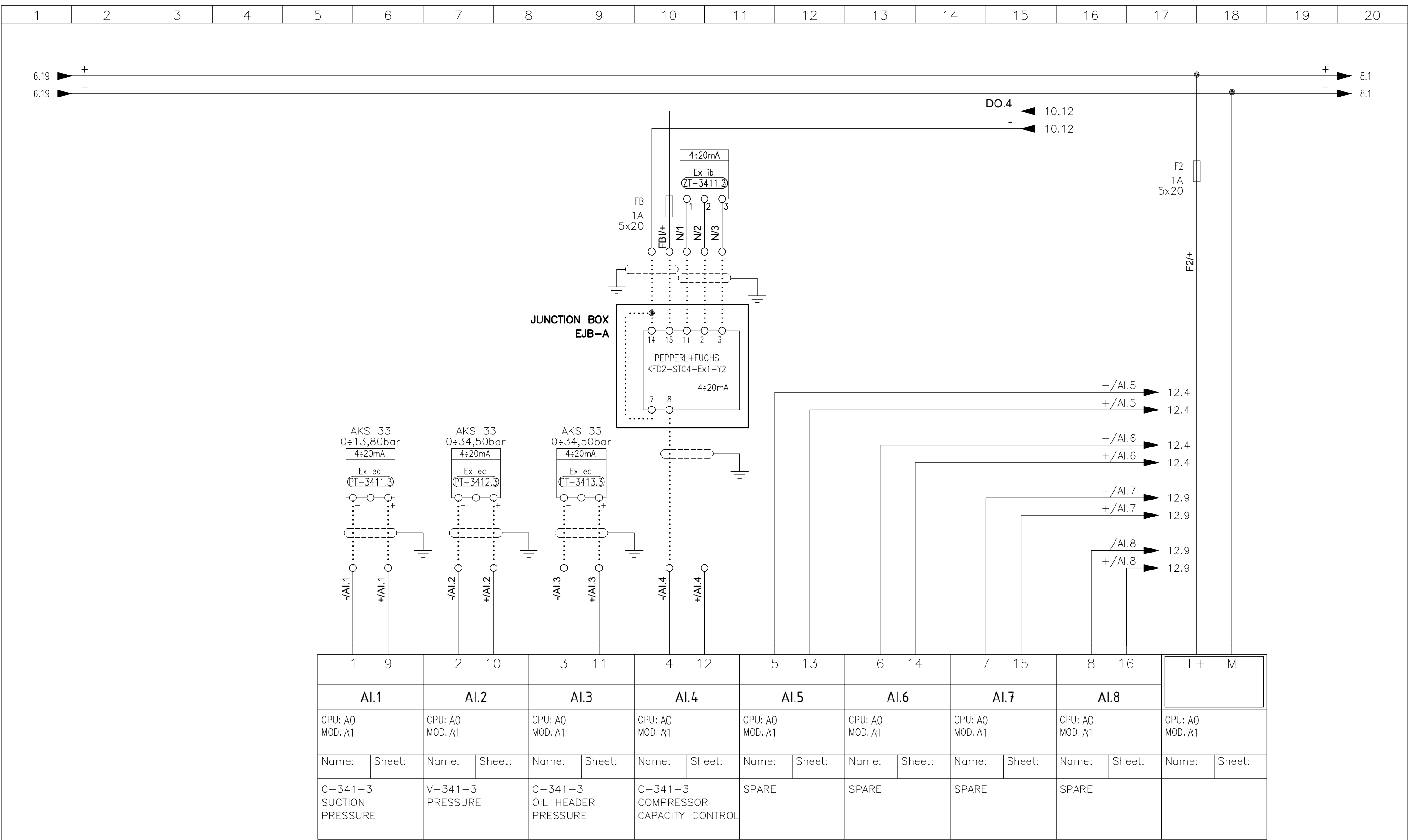


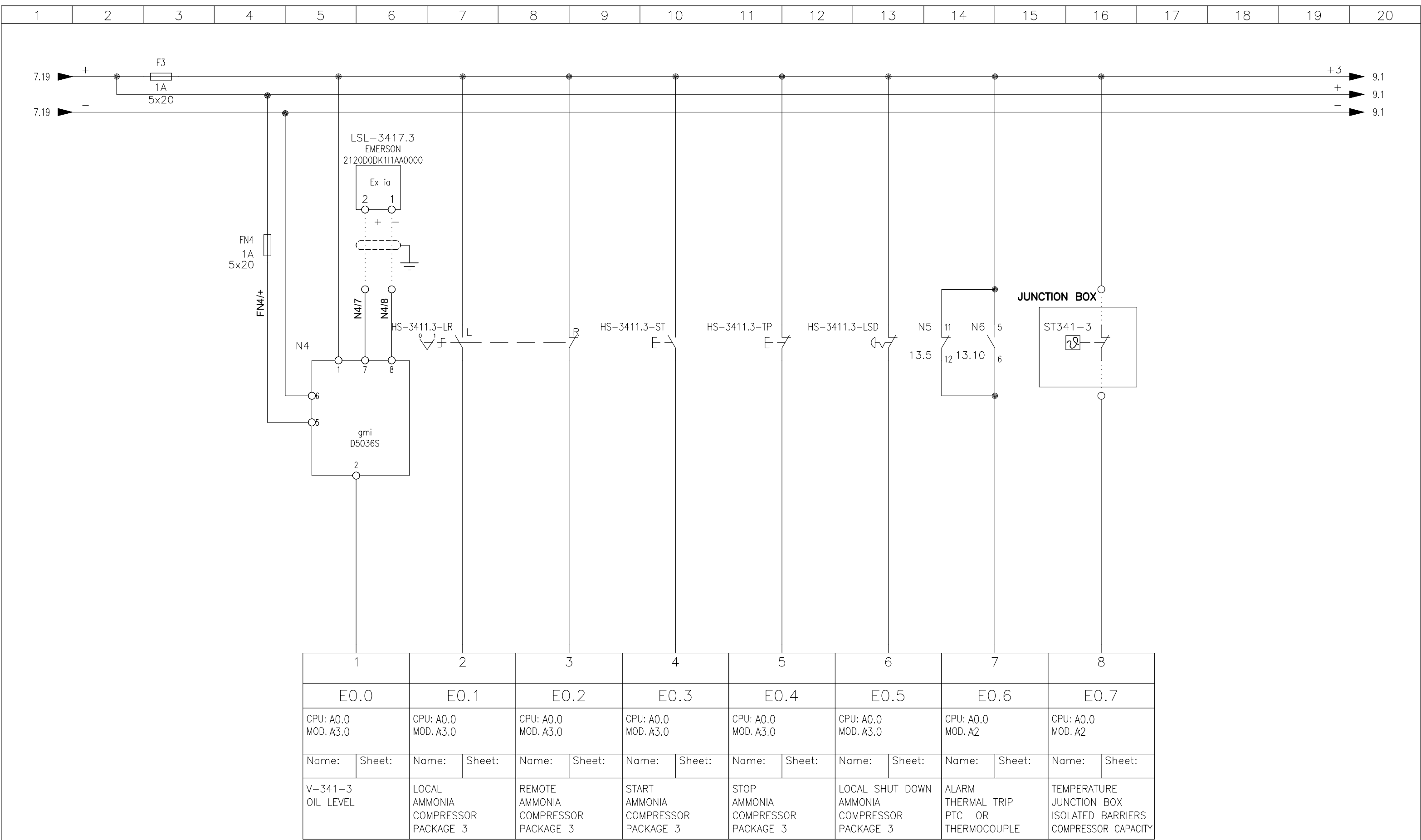


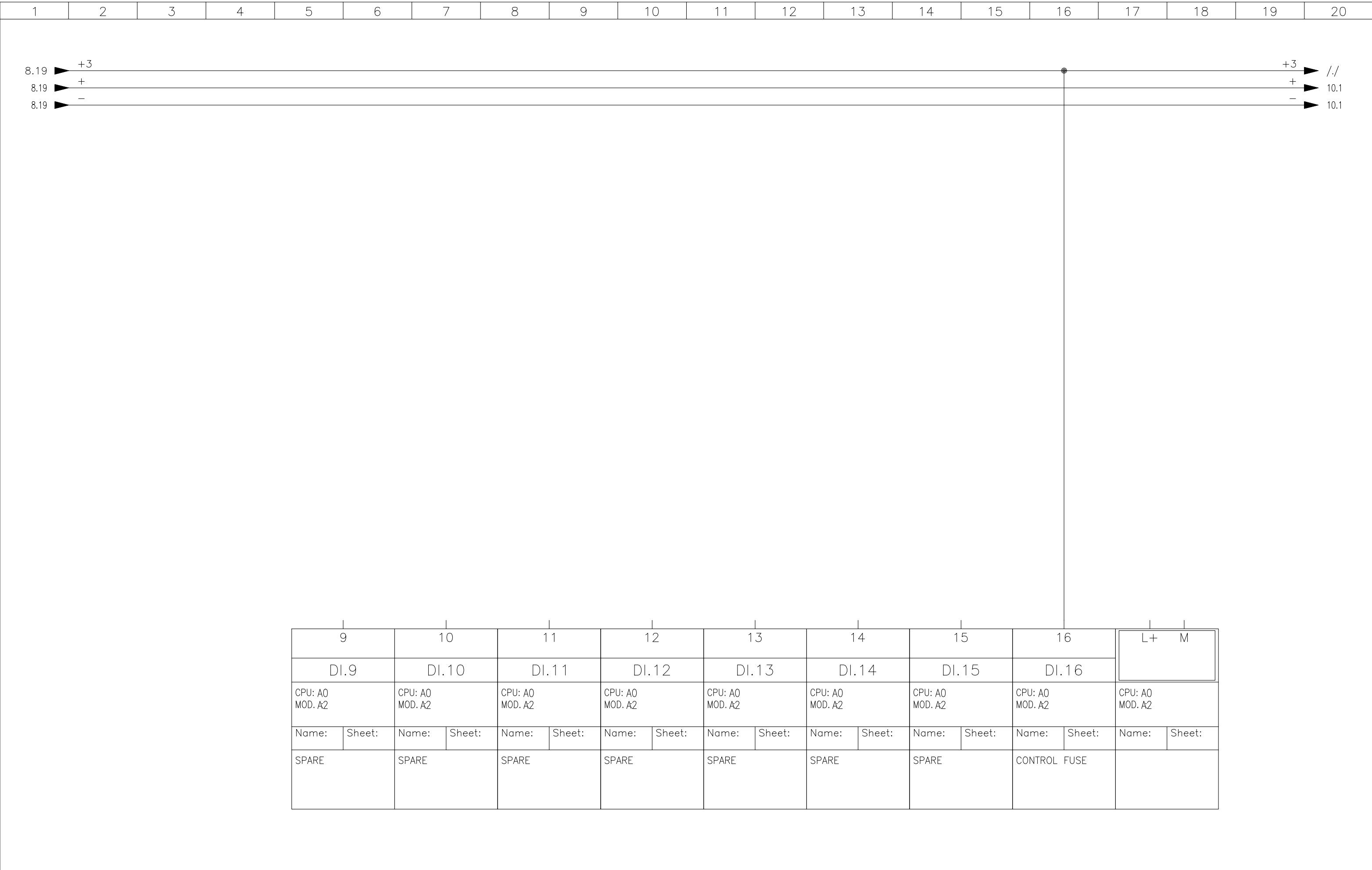
G1: TRIO-PS-2G/1AC/24DC/10 - Power supply unit PHORNIX CONTACT

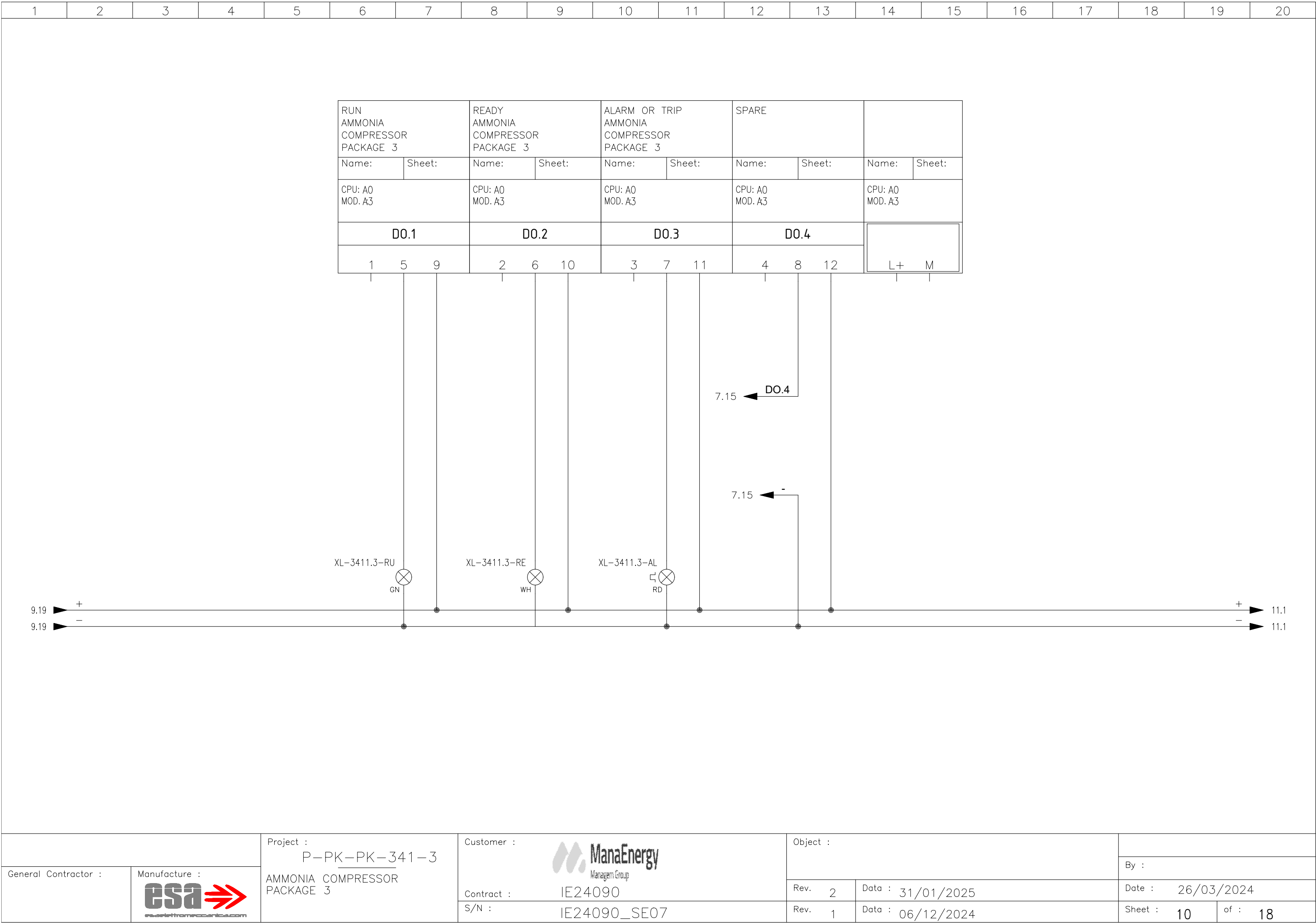
General Contractor : Manufacture : 		Project : P-PK-PK-341-3 AMMONIA COMPRESSOR PACKAGE 3	Customer :  Contract : IE24090 S/N : IE24090_SE07	Object :			
				Rev. 2	Data : 31/01/2025	By :	
				Rev. 1	Data : 06/12/2024	Sheet : 5	of : 18



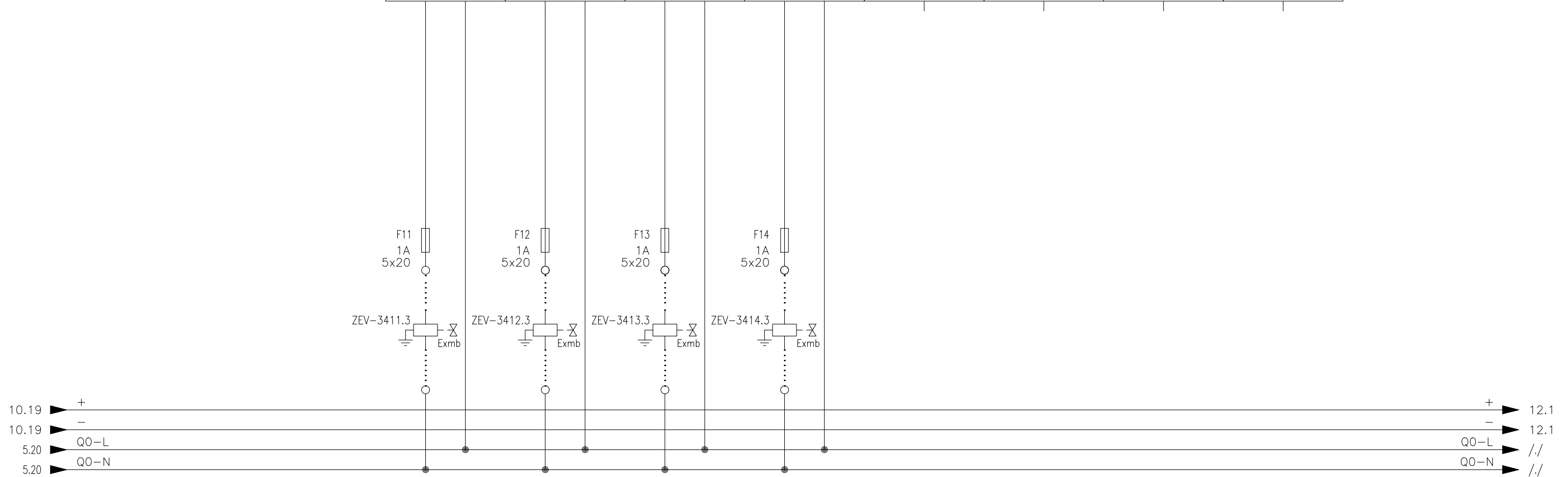


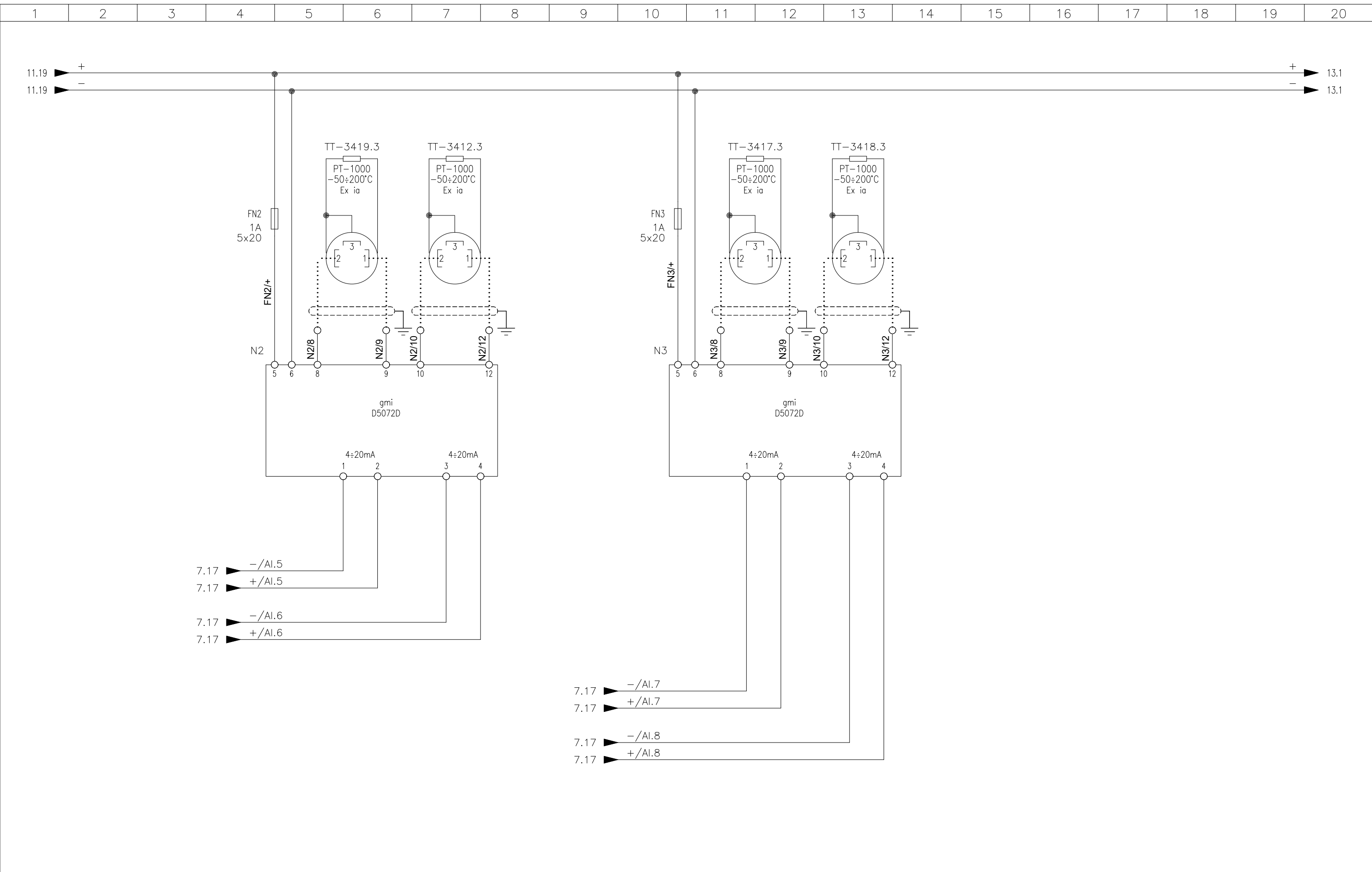






C-341-1 COMPRESSOR CAPACITY CONTROL		C-341-1 COMPRESSOR CAPACITY CONTROL		C-341-1 COMPRESSOR CAPACITY CONTROL		C-341-1 COMPRESSOR CAPACITY CONTROL									
Name:	Sheet:	Name:	Sheet:	Name:	Sheet:	Name:	Sheet:	Name:	Sheet:	Name:	Sheet:	Name:	Sheet:	Name:	Sheet:
CPU: A0 MOD. A4		CPU: A0 MOD. A4		CPU: A0 MOD. A4		CPU: A0 MOD. A4		CPU: A0 MOD. A4		CPU: A0 MOD. A4		CPU: A0 MOD. A4		CPU: A0 MOD. A4	
D0.5		D0.6		D0.7		D0.8									
1	5	2	6	3	7	4	8								





TT-3417.3

PT-1000

-50÷200°C

Ex ia

TT-3418.3

PT-1000

-50÷200°C

Ex ia

FN3

1A

5x20

FN3/+

N3

5

6

8

9

10

12

gmi

D5072D

4÷20mA

1

2

3

4

7.17

-/AI.7

7.17

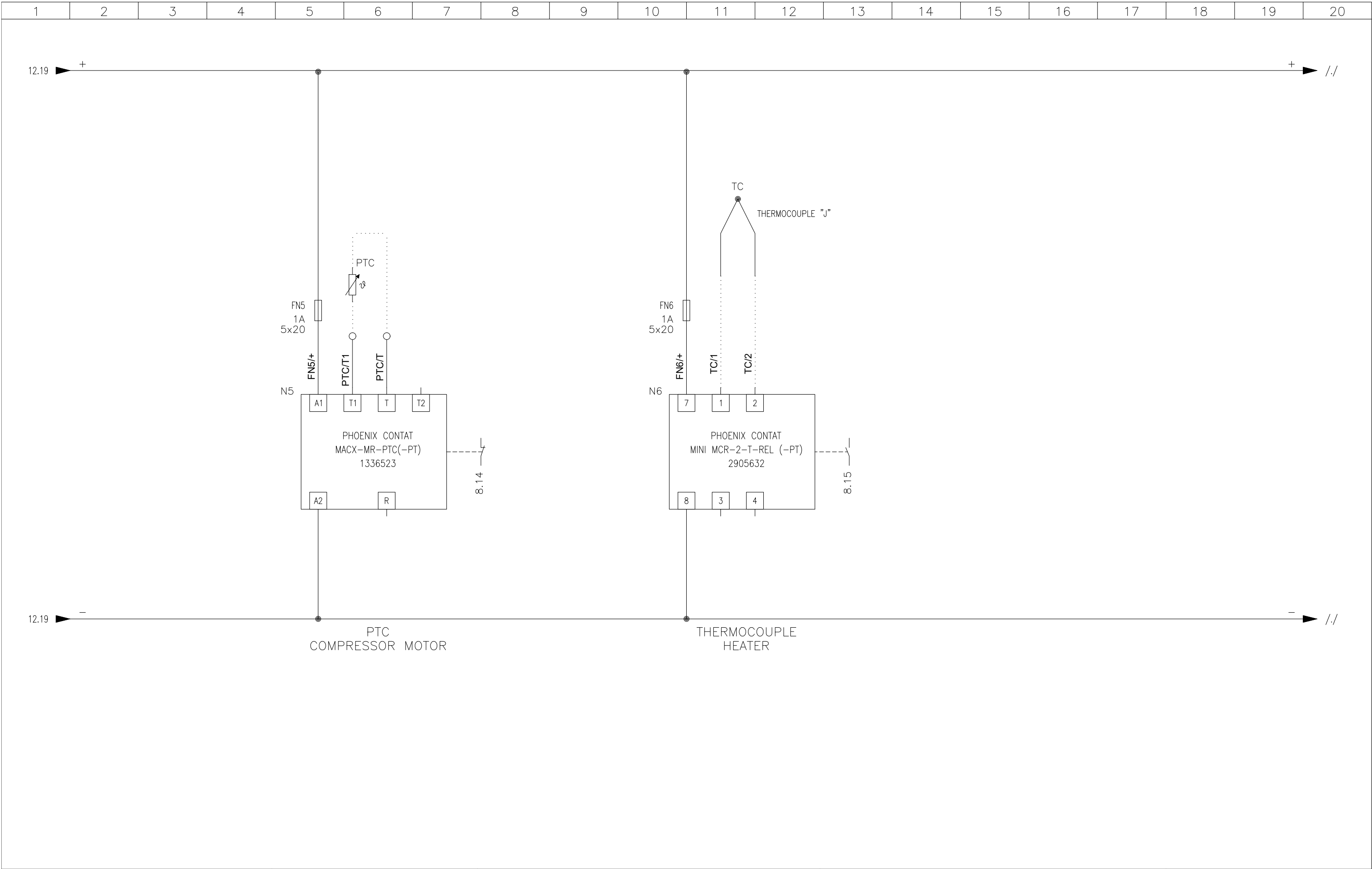
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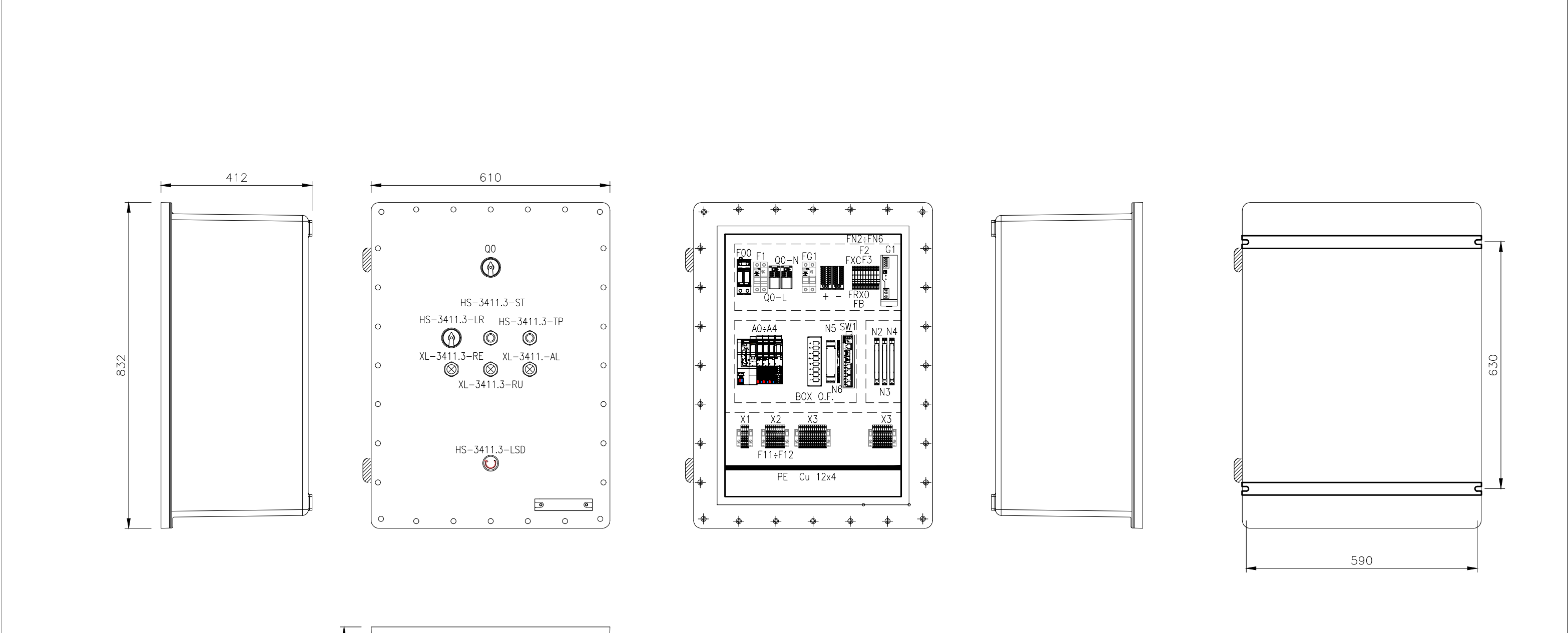
7.17

-/AI.8

7.17

+/AI.8





MILANO-ITALY Tel 024455358 Fax 0248401741

Type: EJB-H
230Vac/50Hz 24Vdc 10A
Max Dis. P: 81 W
Year s.n 25/CMC/0008

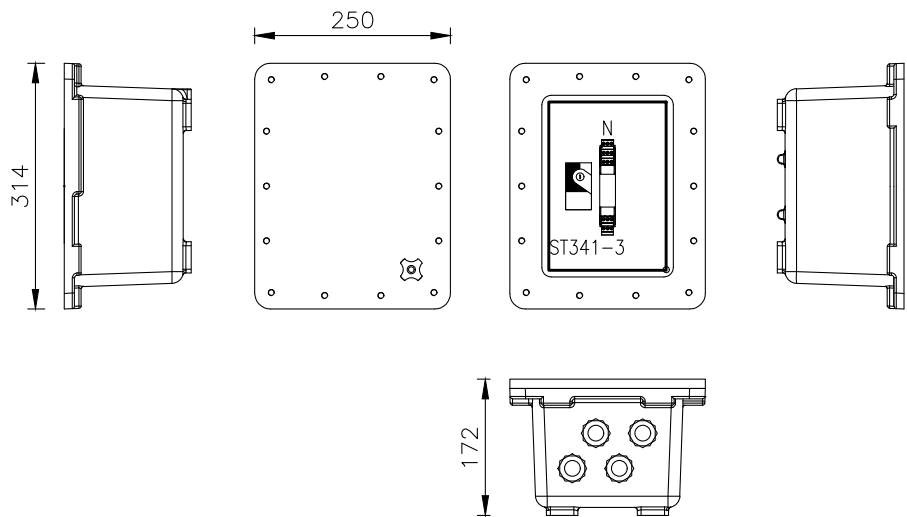
CE 0722 **Ex II 2GD**
INERIS18ATEX0026X/00
Ex db [ia Ga] IIB T6
Ex tb [ia Da] IIC T85 °C IP66
-20°C < Tamb < +50°C

Cable entries: see instructions

WARNINGS: DO NOT OPEN WHILE ENERGIZED
DO NOT OPEN IF AN EXPLOSIVE ATMOSPHERE MAY BE PRESENT
READ INSTRUCTIONS NOTE BEFORE INSTALLATION AND USE

General Contractor :		Manufacture : 	Project : P-PK-PK-341-3 AMMONIA COMPRESSOR PACKAGE 3	Customer : 	Object :		By :		
					Rev. 2	Data : 31/01/2025	Date : 26/03/2024		
		S/N :			IE24090_SE07	Rev. 1	Data : 06/12/2024	Sheet : 14	of : 18

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20								
INITIALS	SUPPLIER PN		DESCRIPTION							SUPPLIER																	
–	3273110		PTFIX 6/18X2,5–NS35 GY – Distribution block							PHOENIX CONTACT																	
+	3273110		PTFIX 6/18X2,5–NS35 GY – Distribution block							PHOENIX CONTACT																	
A0	6AG1155–6AU02–7BN0		ET 200SP, PROFINET interface module IM 155–6PN Standard							SIEMENS																	
A0	6AG1193–6AR00–7AA0		ET 200SP BA 2xRJ45							SIEMENS																	
A1	6AG1134–6GF00–7AA1		AI 8xl 2–/4–wire BA							SIEMENS																	
A1	6AG1193–6BP00–7DA0		ET 200SP BU15–P16+A0+2D push–in terminals							SIEMENS																	
A2	6AG1131–6BH01–7BA0		ET 200SP DI 16x24 V DC							SIEMENS																	
A2	6AG1193–6BP00–7BA0		ET 200SP BU15–P16+A0+2B push–in terminals							SIEMENS																	
A3	6AG1132–6GD51–7BA0		ET 200SP RQ 4x24VDC/2A CO ST							SIEMENS																	
A3	6AG1193–6BP00–7BA0		ET 200SP BU15–P16+A0+2B push–in terminals							SIEMENS																	
A4	6AG1132–6HD01–7BB1		ET 200SP RQ 4x 120V DC..230VAC/5A ST							SIEMENS																	
A4	6AG1193–6BP20–7BB0		ET 200SP BU20–P12+A4+0B push–in terminals							SIEMENS																	
F00	952110		DG M TT 2P 275 – SPD							DEHN																	
F1	2301138		Fuseholder for cylindrical fuses – 1P+N for fuses 10,3x38mm – BCH 1x38N 32A 690V							ITALWEBER																	
F11	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
F12	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
F13	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
F14	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
F2	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
F3	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
FB	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
FG1	2302038		Fuseholder for cylindrical fuses – 2P for fuses 10,3x38mm – BCH 2x38 32A 690V							ITALWEBER																	
FN2	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
FN3	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
FN4	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
FN5	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
FN6	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
FRK0	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
FXC	TBS4F		MODULAR TERMINAL WITH FUSE							CEMBRE																	
G1	2903149		TRIO–PS–2G/1AC/24DC/10 – Power supply unit							PHOENIX CONTACT																	
HS-3411.3-LR	PSRC–D220		CHANGE OVER SWITCH							ITALWEBER																	
HS-3411.3-LSD	PLC–S		MUSHROOM HEAD MAINTAINED (TWIST TO RELEASE)							CMC																	
HS-3411.3-LSD	E20GE11		CONTACT BLOCK N.O. +N.C. WITH FLANGE							ELFI																	
HS-3411.3ST	PLA30		PUSH BUTTON GREEN							CMC																	
HS-3411.3ST	E20GE11		CONTACT BLOCK N.O. +N.C. WITH FLANGE							ELFI																	
HS-3411.3TP	PLA20		PUSH BUTTON RED							CMC																	
HS-3411.3TP	E20GE11		CONTACT BLOCK N.O. +N.C. WITH FLANGE							ELFI																	
N2	D5072D		I.S. SIL2 Multifunction Temperature Converter							GMI																	
N3	D5072D		I.S. SIL2 Multifunction Temperature Converter							GMI																	
N4	D5036S		I.S. SIL2 Switch/Proximity Relay–Out Repeater							GMI																	
N5	1336523		MACX–MR–PTC – Monitoring relay							PHOENIX CONTACT																	
N6	2905632		MINI MCR–2–T–REL – Limit value switch							PHOENIX CONTACT																	
P-PK-341-2	EJB–H		ENCLOSURE 832x612x412 mm L x H x D							CMC																	
Q0	PSRC–320		SELECTOR SWITCH							CMC																	
Q0–L	3273066		PTFIX 6/6X2,5–NS35 GY – Distribution block							PHOENIX CONTACT																	
Q0–N	3273066		PTFIX 6/6X2,5–NS35 GY – Distribution block							PHOENIX CONTACT																	
XL-3411.3	PLD30		PILOT LAMP GREEN							CMC																	
XL-3411.3	O20PD9		LAMPHOLDER BA9S, DIRECT SUPPLY							ELFI																	
XL-3411.3-AL	PLD20		PILOT LAMP RED							CMC																	
XL-3411.3-AL	O20PD9		LAMPHOLDER BA9S, DIRECT SUPPLY							ELFI																	
XL-3411.3-RE	PLD10		PILOT LAMP WHITE							CMC																	
XL-3411.3-RE	O20PD9		LAMPHOLDER BA9S, DIRECT SUPPLY							ELFI																	
General Contractor : Manufacture : 			Project : P–PK–PK–341–3 AMMONIA COMPRESSOR PACKAGE 3					Customer :  Contract : IE24090 S/N : IE24090_SE07					Object :					By :									
													Rev. 2					Data : 31/01/2025					Date : 26/03/2024				
													Rev. 1					Data : 06/12/2024					Sheet : 15		of : 18		



INITIALS	SUPPLIER PN	DESCRIPTION	SUPPLIER
EJB-A	EJB-A	ENCLOSURE 250x314x172 mm L x H x D	CMC
N	KFD2-STC4-Ex1-Y2	SMART Transmitter Power Supply	PEPPERL+FUCHS
ST341-3	TMS NC	THERMOSTAT	O.ERR



MILANO-ITALY Tel 024455358 Fax 0248401741

Type: EJB-A
24Vdc 10A
Max Dis. P: 6 W
Thermal probe: 55±5 °C
Year s.n 25/CMC/0011

CE 0722
II 2GD
INERIS18ATEX0026X/00
Ex db [ia Ga] IIB T6
Ex tb [ia Da] IIIC T85 °C IP66
-20°C < Tamb < +50°C

Cable entries: see instructions

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TYPE OF PROTECTION	
FLAMEPROF ENCLOSURE	EX db
INCREASED SAFETY	EX ec
INTRINSIC SAFETY	EX ia
ENCAPSULATRION	EX mb